

Composed by Yaseen Bilal

Dynamics MTH (404) Notes For Final Term Preparation

Lecture (23)

Projectile Motion

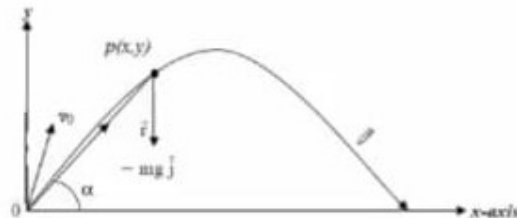
Projectile motion is a form of motion where an object moves in a bilaterally Symmetrical, parabolic path. The path that the object follows is called its trajectory.

Projectile motion only occurs when there is one force applied at the beginning on the

Trajectory, after which the only interference is from gravity.

Trajectory of a Projectile

Let a particle of mass 'm' is projected from a point 'O' with initial velocity ' v_0 ' making an angle ' α ' with horizontal. Take 'O' as origin and horizontal and vertical lines through 'O' as x-axis and y-axis respectively.



$$r = x\hat{i} + y\hat{j}$$

$$\frac{dr}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}$$

$$\vec{v} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}$$

$$\vec{F} = -mg\hat{j}$$

$$F = ma$$

$$-mg\hat{j} = m\left(\frac{d^2x}{dt^2}\hat{i} + \frac{d^2y}{dt^2}\hat{j}\right)$$

$$0\hat{i} - g\hat{j} = \frac{d^2x}{dt^2}\hat{i} + \frac{d^2y}{dt^2}\hat{j}$$

$$\Rightarrow \frac{d^2x}{dt^2} = 0, \quad \frac{d^2y}{dt^2} = g$$

$$\int \frac{d^2x}{dt^2} - \frac{dx}{dt} = A$$

$$\int \frac{d^2y}{dt^2} - \frac{dy}{dt} = -gt + B$$

$$t = 0 \quad \frac{dx}{dt} = V_0 \cos \alpha$$

$$\frac{dy}{dt} = V_0 \sin \alpha$$

$$A = V_0 \cos \alpha \quad B = V_0 \sin \alpha$$

$$\frac{dx}{dt} = (V_0 \cos \alpha) \quad \frac{dy}{dt} = -gt + V_0 \sin \alpha$$

$$x = (V_0 \cos \alpha)t + C$$

$$y = (V_0 \sin \alpha)t - \frac{t^2}{2}g + D$$

$$t = 0, \quad x = 0, \quad y = 0$$

$$C = 0, \quad D = 0$$

$$y = x \tan \alpha - \frac{gx^2}{2V_0^2} \sec^2 \alpha$$

$$x = V_0 \cos \alpha t$$

$$y = V_0 \sin \alpha t - \frac{1}{2}gt^2$$

$$t = \frac{x}{V_0 \cos \alpha}$$

$$y = V_0 \sin \alpha \left(\frac{x}{V_0 \cos \alpha} \right)$$

$$- \frac{1}{2}g \left(\frac{x}{V_0 \cos \alpha} \right)^2$$

Definition:

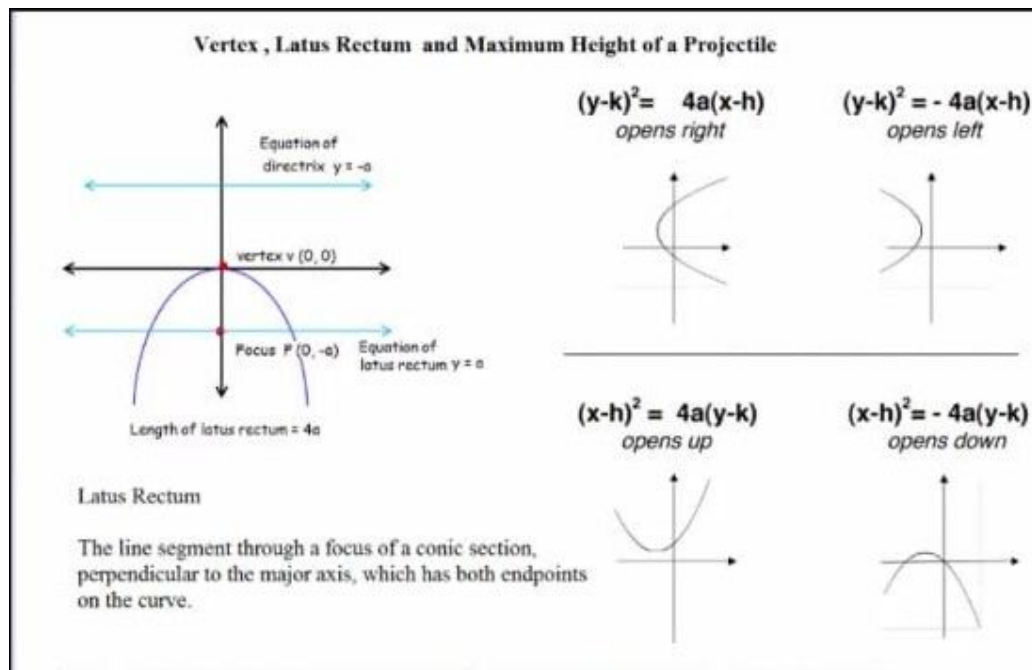
Trajectory is the path described by any projectile.

Lecture (24)

Properties of projectile

Objects experiencing projectile motion have a constant velocity in the horizontal direction, and a constantly changing velocity in the vertical direction. The trajectory resulting from this combination always has the shape of parabola.

Notice that the trajectory is a parabola.



(h, k)

$$y^2 = 4ax$$

$$(x - 4)^2 = 4ay - k$$

$$= -4a(y - k)$$

Latus Rectum

The line through a focus of a conic section,

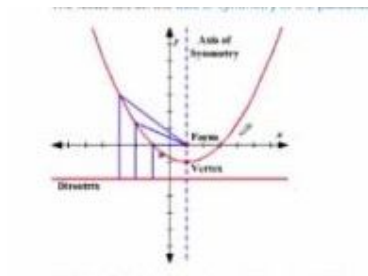
Perpendicular to the major axis, which has both endpoints on the curve.

Focus of a Parabola

A parabola is set of all points in a plane which are at an equal distance away from a given

Point and a given line. The point is called the focus of the parabola and the line is called the directrix.

The focus lies on the axis of symmetry of the parabola.



Vertex, Latus Rectum and Maximum Height of a Projectile

Equation of directrix $y = -a$

vertex $v(0, 0)$

Focus $F(0, -a)$

Equation of latus rectum $y = a$

Length of latus rectum $= 4a$

Latus Rectum

The line segment through a focus of a conic section, perpendicular to the major axis, which has both endpoints on the curve.

Handwritten notes:

- $y^2 = 4ax$
- $x^2 = 4ay$
- $(x-h)^2 = 4a(y-k)$
- $= -4a(y-k)$

Four standard forms of parabola equations and their orientations:

- $(y-k)^2 = 4a(x-h)$ opens right
- $(y-k)^2 = -4a(x-h)$ opens left
- $(x-h)^2 = 4a(y-k)$ opens up
- $(x-h)^2 = -4a(y-k)$ opens down

Finding the focus of a parabola given its equation

if you have the equation of a parabola in vertex form $y = a(z - h)^2 + k$,

then the vertex is a

is at (h, k) and the focus is $\left(h, k + \frac{1}{4a}\right)$.

Properties of Projectile.mp4 - VLC media player

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$$\frac{g x^2}{2 v_0^2} \sec^2 \alpha = x \tan \alpha - y$$

$$x^2 = \left(\frac{2 v_0^2}{g \sec^2 \alpha} \right) x \tan \alpha - y \left(\frac{2 v_0^2}{g \sec^2 \alpha} \right)$$

$$x^2 = \frac{2 \pi v_0^2}{g} \cdot \frac{\sin \alpha}{\cos \alpha} \cdot \cos^2 \alpha - y \frac{2 v_0^2}{g} \cdot \cos^2 \alpha$$

$$= \frac{2 \pi v_0^2 \sin \alpha \cos \alpha}{g} - \frac{2 y v_0^2 \cos^2 \alpha}{g}$$

Properties of Projectile.mp4 - VLC media player

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$$= \frac{2 \pi v_0^2 \sin \alpha \cos \alpha}{g} - \frac{2 y v_0^2 \cos^2 \alpha}{g}$$

$$x^2 - \frac{2 \pi v_0^2 \sin \alpha \cos \alpha}{g} = - \frac{2 y v_0^2 \cos^2 \alpha}{g}$$

$$x^2 - \frac{2 \pi v_0^2 \sin \alpha \cos \alpha}{g} + \left(\frac{v_0^2 \sin \alpha \cos \alpha}{g} \right)^2 = \frac{-2 y v_0^2 \cos^2 \alpha}{g} + \frac{v_0^2 \sin^2 \alpha \cos^2 \alpha}{g}$$

$$\left(x - \frac{v_0^2 \sin \alpha \cos \alpha}{g} \right)^2 = \frac{-2 y v_0^2 \cos^2 \alpha}{g} \left(y - \frac{v_0^2 \sin^2 \alpha}{2g} \right)$$

$$\left(x - \frac{v_0^2 \sin \alpha \cos \alpha}{g}\right)^2 = -\frac{2v_0^2 \cos^2 \alpha}{g} \left(y - \frac{v_0^2 \sin^2 \alpha}{2g}\right)$$

$$(x-h)^2 = 4a(y-k)$$

$$h = \frac{v_0^2 \sin \alpha \cos \alpha}{g} \quad 4a = -\frac{2v_0^2 \cos^2 \alpha}{g}$$

$$k = \frac{v_0^2 \sin^2 \alpha}{2g}$$

$$h = \frac{v_0^2 \sin \alpha \cos \alpha}{g} \quad 4a = -\frac{2v_0^2 \cos^2 \alpha}{g}$$

$$k = \frac{v_0^2 \sin^2 \alpha}{2g}$$

$$\text{Vertex} = (h, k) = \left(\frac{v_0^2 \sin \alpha \cos \alpha}{g}, \frac{v_0^2 \sin^2 \alpha}{2g}\right)$$

$$\text{Latus rectum} = |4a| = \frac{2v_0^2 \cos^2 \alpha}{g}$$

$$\text{Height} = k = \frac{v_0^2 \sin^2 \alpha}{2g}$$

$$\text{Height} = H = \frac{V_0^2 \sin^2 \alpha}{2g}$$

Focus

x-coordinate of Focus = x-coordinate of vertex

$$= \frac{V_0^2 \sin \alpha \cos \alpha}{g}$$

$$= \frac{2 V_0^2 \sin \alpha \cos \alpha}{2g}$$

$$= \frac{V_0^2 \sin 2\alpha}{2g}$$

$$y\text{-coordinate of Focus} = H - \frac{1}{4} (\text{Latus rectum})$$

$$= \frac{V_0^2 \sin^2 \alpha}{2g} - \frac{1}{4} \left(2 \frac{V_0^2 \cos^2 \alpha}{g} \right)$$

$$= \frac{V_0^2 \sin^2 \alpha}{2g} - \frac{V_0^2 \cos^2 \alpha}{2g}$$

$$= -\frac{V_0^2}{2g} (\cos^2 \alpha - \sin^2 \alpha)$$

$$= -\frac{V_0^2}{2g} \cos 2\alpha$$

$$\begin{aligned}
 &= -\frac{v_0^2}{2g} (\cos^2 \alpha - \sin^2 \alpha) \\
 &= -\frac{v_0^2}{2g} \cos 2\alpha
 \end{aligned}$$

Equation of Directrix

$$\begin{aligned}
 y &= H + \frac{1}{4} (\text{Latus rectum}) \\
 &= \frac{v_0^2 \sin^2 \alpha}{2g} + \frac{1}{4} \left(\frac{2v_0^2 \cos^2 \alpha}{g} \right) \\
 &= \frac{v_0^2}{2g} (\sin^2 \alpha + \cos^2 \alpha)
 \end{aligned}$$

$$= \frac{v_0^2}{2g}$$

Time of Flight

$$x = v_0 \cos \alpha t \quad y = (v_0 \sin \alpha)t - \frac{1}{2}gt^2$$

$$y = 0$$

$$(v_0 \sin \alpha)t - \frac{1}{2}gt^2 = 0$$

$$\begin{aligned}
 (v_0 \sin \alpha - \frac{1}{2}gt)t &= 0 \\
 t &= 2v_0 \sin \alpha / g
 \end{aligned}$$

Properties of Projectile.mp4 - VLC media player

Audio Video Subtitle Tools View Help

$$R = (\text{Horizontal velocity}) \times (\text{Time of flight})$$
$$= (V_0 \cos \alpha) \left(\frac{2V_0 \sin \alpha}{g} \right)$$
$$= \frac{V_0^2}{g} \sin 2\alpha$$

Properties of Projectile.mp4 - VLC media player

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$$= \frac{V_0^2}{g} \sin 2\alpha$$
$$\sin 2\alpha = 1$$
$$2\alpha = \sin^{-1}(1)$$
$$2\alpha = 90$$
$$\alpha = 45^\circ$$
$$= \frac{V_0^2}{g} \sin(90)$$
$$R_{\max} = \frac{V_0^2}{g}$$

Examples of projectile Motion -1

Question

Determine the maximum possible range for a projectile fired from a cannon having

muzzle velocity v_0 and prove that the height reached in this case is

$$\frac{v_0^2}{4g}$$

Solution:

Range = horizontal velocity $\times$ Time of flight

$$= v_0 \cos \alpha \times \frac{2v_0 \sin \alpha}{g}$$

$$= \frac{v_0^2 \sin 2\alpha}{g}$$

$\sin 2\alpha$ is max

$$\sin 2\alpha = 1$$

$$2\alpha = \sin^{-1}(1)$$

$$2\alpha = 90$$

$$\alpha = 45$$

$$Range = \frac{v_0^2}{g}$$

$$Range_{\max} = \frac{v_0^2}{g}$$

$$v_0 = 1 \text{ mile / sec} = 1760 \times 3 / \text{sec}$$

$$= 5280 \text{ mile / sec}$$

$$R_{\max} = (5280)^2 \quad g = 9.8 \text{ m / sec}^2$$
$$= 32$$

Lecture (26,27,28,29,33) are included

Examples of Projectile Motion –(II),(III),(IV),(V),(VII)

Question

A projectile having horizontal range T , reaches a maximum height H.

Prove

that it must have been launched d with

a) an initial speed equal to

$$\sqrt{\frac{g(R^2 + 16H^2)}{8H}}$$

b) at an angle with horizontal given by

$$\sin^{-1}\left(\frac{4H}{\sqrt{R^2 + 16H^2}}\right)$$

Solution:

Let V_0 be the velocity of projection and α is the angle of projection.

$$R = \frac{v_o^2}{g} \sin 2\alpha$$

$$\Rightarrow R^2 = \frac{4v_o^4 \sin^2 \alpha \cos^2 \alpha}{g^2} \dots\dots\dots(i)$$

$$\text{Given } H \text{ is the maximum height of the projectile. } H = \frac{V_o^2 \sin^2 \alpha}{2g} \dots\dots\dots(ii)$$

$$\text{Divide (i) and (ii)} \quad \frac{R^2}{H} = \frac{4v_o^2 \sin^2 \alpha \cos^2 \alpha}{g^2} \times \frac{2g}{v_o^2 \sin^2 \alpha}$$

$$\Rightarrow \frac{R^2}{H} = \frac{8V_o^2 \cos^2 \alpha}{g}$$

$$\frac{gR^2}{8H} = V_o^2 \cos^2 \alpha \dots\dots\dots(iii)$$

$$\text{From equation (ii)} \quad 2gH = V_o^2 \sin^2 \alpha \dots\dots\dots(iv)$$

$$\text{Adding equation (iii) and (iv)} \quad \frac{gR^2}{8H} + 2gH = V_o^2 \cos^2 \alpha + V_o^2 \sin^2 \alpha$$

$$V_o^2 = \frac{gR^2 + 16H^2}{8H} \dots\dots\dots(v)$$

$$\text{Thus, } V_o = \sqrt{\frac{gR^2 + 16H^2}{8H}}$$

Now, we find the angle ' α '.

$$\text{From equation (ii),} \quad \sin^2 \alpha = \frac{2gH}{V_o^2}$$

$$\Rightarrow \sin^2 \alpha = \frac{2gH}{\frac{g(R^2 + 16H^2)}{8H}} = \frac{2gH \times 8H}{g(R^2 + 16H^2)}$$

$$\Rightarrow \sin^2 \alpha = \frac{16H^2}{R^2 + 16H^2}$$

$$\Rightarrow \sin \alpha = \frac{4H}{\sqrt{R^2 + 16H^2}}$$

$$\text{Thus, } \alpha = \sin^{-1} \left(\frac{4H}{\sqrt{R^2 + 16H^2}} \right)$$

Example:

$$\text{Thus, } [AD] = 4 \cos \alpha \sqrt{H(H \sin^2 \alpha - h)}$$

Q.14. A shell fired with speed V at an elevation θ , hits an airship at height H , which is moving horizontally away from the gun with speed V_0 . Show that if $(2V \cos \theta - V_0) (V^2 \sin^2 \theta - 2gH)^{1/2} = V_0 V \sin \theta$, the shell might also have hit the airship if the latter had remained stationary in the position it occupied when the gun was actually fired.-

Solution:

Let 'A' be the position of the airship when the gun was actually fired. Let the shell hit the airship at point 'B'. If the airship remains stationary at point 'A', then the shell also hits the airship at 'A'.

Given that the shell is fired with speed V at an elevation θ .

$$\text{Now, } y = (V \sin \theta) t - \frac{1}{2} g t^2$$

$$\text{But } y = H \quad \text{Therefore, } H = (V \sin \theta) t - \frac{1}{2} g t^2$$

$$\text{Multiply by '2' } 2H = (2V \sin \theta) t - g t^2$$

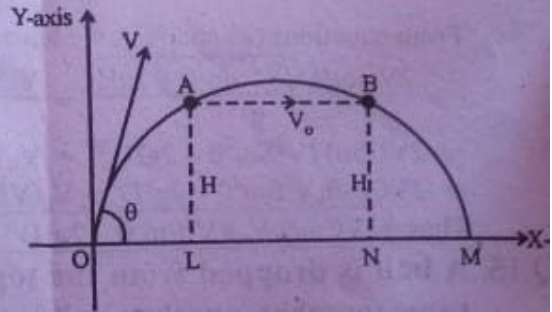
$$\Rightarrow g t^2 - (2V \sin \theta) t + 2H = 0$$

$$t = \frac{2V \sin \theta \pm \sqrt{4V^2 \sin^2 \theta - 4(g)(2H)}}{2g}$$

$$\text{So, } t = \frac{V \sin \theta \pm \sqrt{V^2 \sin^2 \theta - 2gH}}{g}$$

$$\text{Thus, } t_1 = \frac{V \sin \theta - \sqrt{V^2 \sin^2 \theta - 2gH}}{g} \text{ is the time of motion of shell from 'O' to 'A'}$$

$$t_2 = \frac{V \sin \theta + \sqrt{V^2 \sin^2 \theta - 2gH}}{g} \text{ is the time of motion of shell from 'O' to 'B'}$$



Thus, $t_2 - t_1 = \frac{2\sqrt{V^2 \sin^2 \theta - 2gH}}{g}$ (i)

This is the time for shell to move from 'A' to 'B'.

Now, the horizontal distance covered by the shell in time $t_2 - t_1$ with uniform horizontal speed $V \cos \theta$ is

$$|AB| = (V \cos \theta)(t_2 - t_1) \text{ (using } S = vt)$$

Or $|AB| = \frac{2V \cos \theta \sqrt{V^2 \sin^2 \theta - 2gH}}{g}$ (ii)

Now, the shell will hit the airship at point 'B' if the time taken by the ship to move from 'A' to 'B' is equal to the time for shell to move from O to B.

So, the time for airship to move from A to B is also t_2 given by

$$t_2 = \frac{V \sin \theta + \sqrt{V^2 \sin^2 \theta - 2gH}}{g}$$

Now, $|AB| = V_o t_2$ (for ship to move from A to B)

So, $|AB| = \frac{V_o V \sin \theta + V_o \sqrt{V^2 \sin^2 \theta - 2gH}}{g}$ (iii)

From equations (ii) and (iii), we have

$$\frac{2V \cos \theta \sqrt{V^2 \sin^2 \theta - 2gH}}{g} = \frac{V_o V \sin \theta + V_o \sqrt{V^2 \sin^2 \theta - 2gH}}{g}$$

$$\Rightarrow 2V \cos \theta (V^2 \sin^2 \theta - 2gH)^{1/2} = V_o V \sin \theta + V_o (V^2 \sin^2 \theta - 2gH)^{1/2}$$

$$\text{or } 2V \cos \theta (V^2 \sin^2 \theta - 2gH)^{1/2} - V_o (V^2 \sin^2 \theta - 2gH)^{1/2} = V_o V \sin \theta$$

$$\text{Thus, } (2V \cos \theta - V_o)(V^2 \sin^2 \theta - 2gH)^{1/2} = V_o V \sin \theta$$

Example:

Q.18. An aeroplane is flying with uniform speed V_0 in an arc of a vertical circle of radius 'a' whose centre is at a height h vertically above a point O of the ground. If a bomb is dropped from the aeroplane when at a height Y and strikes the ground at O, show that Y satisfies the equation

$$ky^2 + y(a^2 - 2hk) + k(h^2 - a^2) = 0, \text{ where } k = h + \frac{ga^2}{2V_0^2}$$

Solution:

Let the bomb is dropped from point O' (origin). Also let the initial velocity of bomb is V_0 and angle of projection is ' $-\alpha$ ' with the horizontal. So, the equation of the path of projectile is

$$y = x \tan(-\alpha) - \frac{gx^2 \sec^2(-\alpha)}{2V_0^2}$$

$$\Rightarrow y = -x \tan \alpha - \frac{gx^2 \sec^2 \alpha}{2V_0^2}$$

Since the bomb hits at O(x, -y), so it will satisfy above equation.

$$-y = -x \tan \alpha - \frac{gx^2 \sec^2 \alpha}{2V_0^2}$$

$$\Rightarrow y = x \tan \alpha + \frac{gx^2 \sec^2 \alpha}{2V_0^2} \dots (i)$$

For $\Delta O'AC$,

$$|O'A|^2 = |O'C|^2 - |AC|^2$$

$$\Rightarrow x^2 = a^2 - (h-y)^2$$

$$\text{or } x = \sqrt{a^2 - (h-y)^2}$$

$$\text{Further, } \tan \alpha = \frac{\sqrt{a^2 - (h-y)^2}}{h-y}$$

$$\cos \alpha = \frac{h-y}{a} \Rightarrow \sec \alpha = \frac{a}{h-y}$$

Putting these values in equation (i)

$$Y = \sqrt{a^2 - (h-y)^2} \left(\frac{\sqrt{a^2 - (h-y)^2}}{h-y} \right) + \frac{g}{2V_0^2} [a^2 - (h-y)^2] \cdot \frac{a^2}{(h-y)^2}$$

Multiply by $(h-y)^2$

$$y(h-y)^2 = [a^2 - (h-y)^2](h-y) + \frac{ga^2}{2V_0^2} [a^2 - (h-y)^2]$$

$$y(h^2 - 2hy + y^2) = (a^2 - h^2 - y^2 + 2hy)(h-y)$$

$$+ \frac{ga^2}{2V_0^2} [a^2 - h^2 - y^2 + 2hy]$$

$$yh^2 - 2hy^2 + y^3 = ha^2 - h^3 - hy^2 + 2h^2y - a^2y + h^2y + y^3 - 2hy^2$$

$$+ \frac{ga^2(a^2)}{2V_0^2} - \frac{ga^2h^2}{2V_0^2} - \frac{ga^2y^2}{2V_0^2} + \frac{ga^2hy}{V_0^2}$$

$$0 = -\left(h + \frac{ga^2}{2V_0^2}\right)y^2 + \left(-a^2 + 2h^2 + \frac{ga^2h}{V_0^2}\right)y + ha^2 - h^3 + \frac{ga^2(a^2)}{2V_0^2} - \frac{ga^2h^2}{2V_0^2}$$

$$-y^2\left(h + \frac{ga^2}{2V_0^2}\right) + \left[2h\left(h + \frac{ga^2}{2V_0^2}\right) - a^2\right]y + \left[h(a^2 - h^2) + \frac{ga^2}{2V_0^2}(a^2 - h^2)\right] = 0$$

$$-y^2\left(h + \frac{ga^2}{2V_0^2}\right) + \left\{2h\left(h + \frac{ga^2}{2V_0^2}\right) - a^2\right\}y + \left\{(a^2 - h^2)\left(h + \frac{ga^2}{2V_0^2}\right)\right\} = 0$$

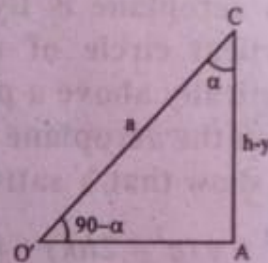
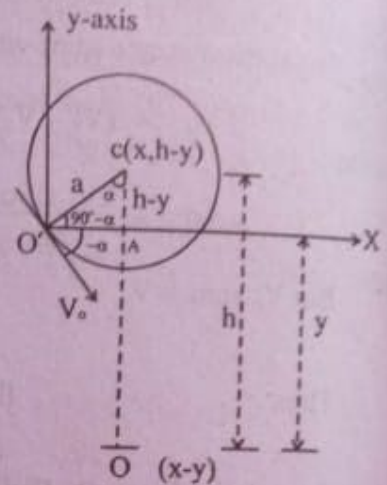
$$\text{Put } h + \frac{ga^2}{2V_0^2} = k$$

$$\text{So, } -y^2k + (2hk - a^2)y + \{(a^2 - h^2)k\} = 0$$

$$\text{Or } +y^2k - (2hk - a^2)y - (a^2 - h^2)k = 0$$

$$\text{Or } y^2k + (a^2 - 2hk)y + (h^2 - a^2)k = 0$$

Is required.



Lecture (30)

Speed of Projectile Examples

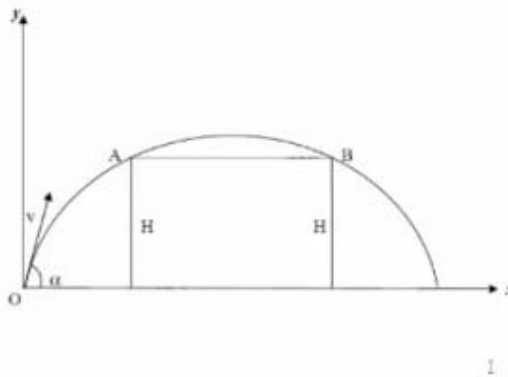
Question:

A shell fired with speed V at an elevation θ , hits an airship at height H ...
Which is moving

horizontally away from the gun with speed V_o . Show that if

$$(2V \cos \theta - v_o) \sqrt{(v^2 \sin^2 \theta - 2gH)} = v_o V \sin \theta$$

The shell might also have hit the air ship if the latter had remained stationary in the position it occupied when the gun was actually fired.



Solution:

$$t = \frac{2V \sin \theta \pm \sqrt{4V^2 \sin^2 \theta - 4(g)(2H)}}{2g}$$

$$\text{So, } t = \frac{V \sin \theta \pm \sqrt{V^2 \sin^2 \theta - 2gH}}{g}$$

$$\text{Thus, } t_1 = \frac{V \sin \theta \pm \sqrt{V^2 \sin^2 \theta - 2gH}}{g} \text{ is the time of motion of shell from 'O' to 'A'}$$

$$t_2 = \frac{V \sin \theta \pm \sqrt{V^2 \sin^2 \theta - 2gH}}{g} \text{ is the time of motion of shell from 'O' to 'B'}$$

$$\text{Thus, } t_2 - t_1 = \frac{2\sqrt{V^2 \sin^2 \theta - 2gH}}{g} \dots\dots\dots(1)$$

This is the time for shell to move from 'A' to 'B'

Now, the horizontal distance covered by the shell in time $t_2 - t_1$ with uniform horizontal speed $V \cos \theta$ is

$$|AB| = (V \cos \theta)(t_2 - t_1) \quad (u \sin g \quad S = vt)$$

$$\text{or } |AB| = \frac{2V \cos \theta \sqrt{V^2 \sin^2 \theta - 2gH}}{g} \dots\dots\dots(2)$$

Now, the shell will hit the airship at point 'B' if the time taken by the ship to move from 'A' to 'B' is equal to the time for shell to move from O to B.

So, the time for airship to move from A to B is also t_2 given by

$$t_2 = \frac{V \sin \theta \pm \sqrt{V^2 \sin^2 \theta - 2gH}}{g}$$

Now, $|AB| = V_0 t_2$ (for ship to move from A to B)

$$\text{So, } |AB| = \frac{V_0 \sin \theta + V_0 \sqrt{V^2 \sin^2 \theta - 2gH}}{g} \dots \dots \dots (3)$$

From equations (ii) and (iii), we have

$$\frac{2V \cos \theta + V_0 \sqrt{V^2 \sin^2 \theta - 2gH}}{g} = \frac{V_0 \sin \theta + V_0 \sqrt{V^2 \sin^2 \theta - 2gH}}{g}$$

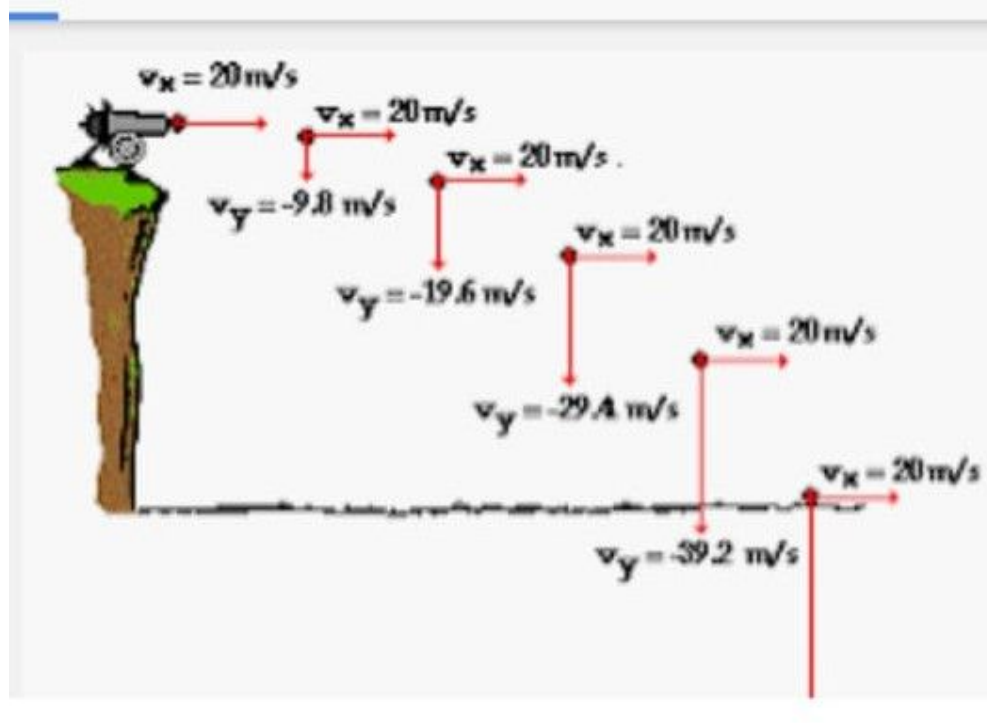
$$\Rightarrow 2V \cos \theta (V^2 \sin^2 \theta - 2gH)^{\frac{1}{2}} = V_0 \sin \theta + V_0 (V^2 \sin^2 \theta - 2gH)^{\frac{1}{2}}$$

$$\text{or } 2V \cos \theta (V^2 \sin^2 \theta - 2gH)^{\frac{1}{2}} - V_0 (V^2 \sin^2 \theta - 2gH)^{\frac{1}{2}} = V_0 V \sin \theta$$

Thus,

$$(2V \cos \theta - V_0) (V^2 \sin^2 \theta - 2gH)^{\frac{1}{2}} = V_0 V \sin \theta$$

Example



The vertical velocity of a projectile changes by 9.8m/s each second, The horizontal motion of a projectile is independent of its vertical motion.

Time	Horizontal Velocity	Vertical Velocity
0 s	73.1 m/s, Right	19.8m/s, up
1 s	73.1 m/s, Right	9.8m/s, up
2 s	73.1 m/s, Right	0m/s
3 s	73.1 m/s, Right	9.8 m/s, down

Example

For example, you throw the ball straight upward, or you kick a ball and give it a speed at

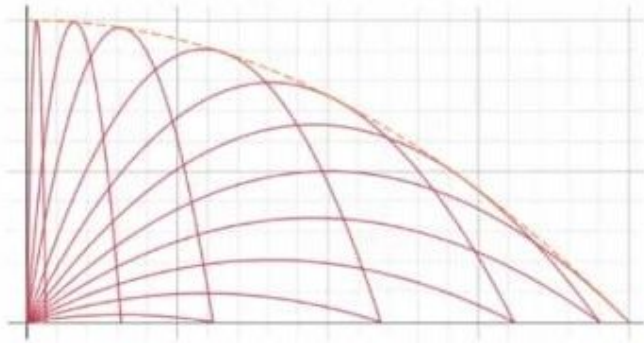
an angle to the horizontal or you just drop things and make them free fall; all these are examples of projectile motion. In projectile motion, gravity is the only force acting on the object.

Lecture (31)

Parabola of Safety

For a given launch velocity v and launch angle θ

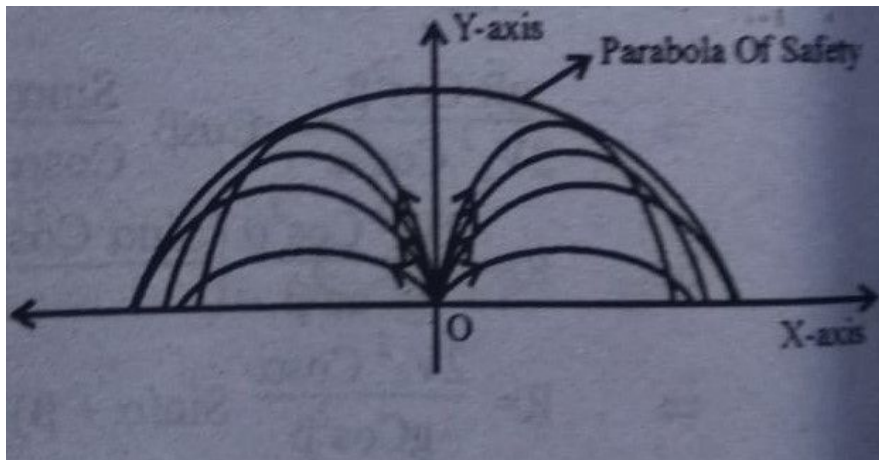
The parabola of safety or safety parabola is the envelope of the parabolic trajectories of projectiles



Parabola of Safety:

In classical mechanics and ballistics, the parabola of safety or safety parabola is the envelope of the parabolic trajectories of projectiles shot from a certain point with a given speed at different angles to horizon in a fixed vertical plane.

Define safety of parabola and derive its equation.



The parabola of safety is the boundary curve in a vertical plane which include all possible paths of projectile with the same (a given) initial velocity.

The maximum range of a projectile having initial velocity V_o , on an inclined plane making an angle β with horizontal is

$$R = \frac{V_o^2}{g(1 + \sin \beta)}$$

$$\text{or } R(1 + \sin \beta) = \frac{V_o^2}{g}$$

$$R + R \sin \beta = \frac{V_o^2}{g} \quad (\text{But } R \sin \beta = y)$$

$$\Rightarrow R + y = \frac{V_o^2}{g}$$

$$\text{or } R = \frac{V_o^2}{g} - y \quad \text{squaring both sides, } R^2 = \left(\frac{V_o^2}{g} - y \right)^2$$

$$\Rightarrow x^2 + y^2 = \frac{V_o^4}{g^2} + y^2 - \frac{2V_o^2 y}{g} \quad (\because R^2 = x^2 + y^2)$$

$$\text{So, } x^2 = \frac{V_o^4}{g^2} - \frac{2V_o^2 y}{g}$$

$$\text{or } x^2 = -\frac{2V_o^2 y}{g} + \frac{V_o^4}{g^2}$$

$$\Rightarrow (x - 0)^2 = \frac{-2V_o^2 y}{g} \left(y - \frac{V_o^2}{2g} \right)$$

This is the equation of the parabola of safety.

$$(1) \quad \text{vertex } v \left(0, \frac{V_o^2}{2g} \right)$$

$$(2) \quad \text{Length of latus rectum} = \frac{V_o^2}{2g}$$

$$(3) \text{Focus } S(0,0)$$

Lecture (32)

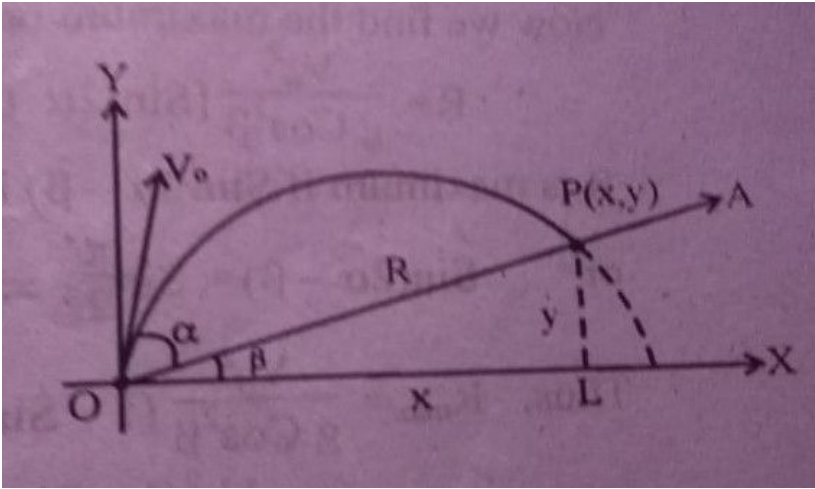
Range on Inclined Plane

Question:

A particle is projected in a vertical plane with velocity V_o making an angle α with horizontal line. Find the maximum range of projectile and the time of flight on an inclined plane making an angle β ($\beta < \alpha$) with the horizontal line.

Solution:

Given that the particle is projected
with velocity V_0 making an angle ' α '



with horizontal. During its flight, the
particle hits the inclined plane OA at
point P(x,y) making an angle ' β ' with
horizontal.

But $x = R \cos \beta$ $y = R \sin \beta$

So, coordinates of P are $(R \cos \beta, R \sin \beta)$.

As $p(R \cos \beta, R \sin \beta)$ lies on the path of parabola whose equation is

$$y = x \tan \alpha - \frac{gx^2 \sec^2 \alpha}{2V_o^2}$$

So, we have

$$R \sin \beta = R \cos \beta \tan \alpha - \frac{gR^2 \cos^2 \beta \sec^2 \alpha}{2V_o^2}$$

$$\Rightarrow \sin \beta = \cos \beta \tan \alpha - \frac{gR^2 \cos^2 \beta \sec^2 \alpha}{2V_o^2}$$

$$\Rightarrow \frac{gR^2 \cos^2 \beta}{2V_o^2 \cos^2 \alpha} = \cos \frac{\sin \alpha}{\cos \alpha} - \sin \beta$$

$$\frac{gR^2 \cos^2 \beta}{2V_o^2 \cos^2 \alpha} = \frac{\sin \alpha \cos \beta - \cos \alpha \sin \beta}{\cos \alpha}$$

$$\Rightarrow R = \frac{2V_o^2 \cos \alpha}{g \cos^2 \beta} \sin(\alpha - \beta)$$

$$\text{or } R = \frac{V_o^2}{g \cos^2 \beta} [2 \sin(\alpha - \beta) \cos \alpha] \dots \dots \dots (i)$$

$$R = \frac{V_o^2}{g \cos^2 \beta} [2 \sin(\alpha - \beta + \alpha) + \sin(\alpha - \beta - \alpha)]$$

$$\text{Thus } R = \frac{V_o^2}{g \cos^2 \beta} [\sin(2\alpha - \beta) - \sin \beta]$$

Now, in order to find the time of flight 'T', we use the equation.

$$x = (V_o \cos \alpha) t$$

$$\text{So, } R \cos \beta = (V_o \cos \alpha) T$$

$$\Rightarrow T = \frac{R \cos \beta}{(V_o \cos \alpha)}$$

$$\text{Using equation (i), } T = \frac{\cos \beta}{(V_o \cos \alpha)} \left(\frac{V_o^2}{g \cos^2 \beta} \{2 \sin(\alpha - \beta) \cos \alpha\} \right)$$

$$\text{Thus, } T = \frac{2V_o}{g \cos \beta} \sin(\alpha - \beta)$$

Now, we find the maximum range. Take

$$R = \frac{V_o^2}{g \cos^2 \beta} [\sin(2\alpha - \beta) - \sin \beta]$$

$$R \text{ is maximum if } \sin(2\alpha - \beta) \text{ is maximum} \quad \text{i.e.,} \quad \sin(2\alpha - \beta) = 1$$

$$\text{or } \sin(2\alpha - \beta) = \sin \frac{\pi}{2} \Rightarrow 2\alpha - \beta = \frac{\pi}{2}$$

$$\text{Thus } R_{\max} = \frac{V_o^2}{g \cos^2 \beta} (1 - \sin \beta)$$

$$R_{\max} = \frac{V_o^2 (1 - \sin \beta)}{g (1 + \sin \beta) (1 - \sin \beta)}$$

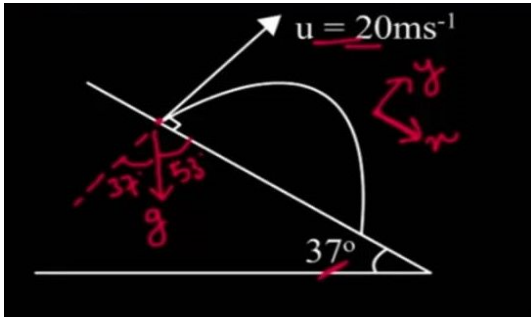
$$\text{Thus, } R_{\max} = \frac{V_o^2}{g (1 + \sin \beta)}$$

This is the maximum range.

Question:

Find the range of the projectile on the inclined plane which is projected perpendicular

to the inclined plane with velocity 20 m/s as shown in figure.



Solution:

$$U_x = 0$$

$$U_y = 20 \text{ m/s}$$

$$a_x = g \sin 37 = \frac{3}{5} g$$

$$a_y = g \cos 37 = \frac{4}{5} g$$

$$5 = 4y t + \frac{1}{2} a_y t^2$$

$$0 = 20t + \frac{1}{2} \times (-g \cos 37) \times t^2$$

$$20t = \frac{1}{2} \times 10 \times \frac{4}{5} \times t^2$$

$$20t = 4t^2$$

$$t = 0, 5 \text{ s}$$

$$5_x = U_x t + \frac{1}{2} a_x t^2$$

$$= \frac{1}{2} \times 10 \times \frac{3}{5} \times 20^5$$

$$= 7.5 \text{ m}$$

Lecture (34)

Introduction to orbital motion

Central Force:

If a particle is moving in an orbit under the influence of a force whose line of action passes through some fixed point, then such a force is called central force and the fixed point is called its centre. The central force may be attractive or repulsive.

THEOREM: The orbit of a particle moving under a central force is necessarily a plane curve.

Proof:

Let $\vec{F}$ be the central force acting on a particle of mass "m" and the origin "O" be the centre of

$\vec{F}$ as shown in the figure.

Let $p(\vec{r})$ be the position of the particle at any time t.

So, $\vec{OP} = \vec{r}$

Since, $\vec{F}$ and $\vec{r}$ are along the same line,

So, $\vec{r} \times \vec{F} = 0$. But $\vec{F} = m\vec{a}$

$$\text{or } \vec{F} = m \frac{d\vec{v}}{dt} \quad \left(\because \vec{a} = \frac{d\vec{v}}{dt} \right)$$

$$\vec{r} \times m \frac{d\vec{v}}{dt} = 0$$

$$\Rightarrow \vec{r} \times \frac{d\vec{v}}{dt} = 0 \dots \dots \dots (i)$$

Now, $\vec{v} \times \vec{v} = 0$

$$\text{or } \frac{d\vec{r}}{dt} \times \vec{v} = 0 \dots \dots \dots (ii) \quad \left(\because \vec{v} = \frac{d\vec{r}}{dt} \right)$$

Adding equation (i) and (ii), we have

$$\vec{r} \times \frac{d\vec{v}}{dt} + \frac{d\vec{r}}{dt} \times \vec{v} = 0$$

$$\Rightarrow \frac{d}{dt}(\vec{r} \times \vec{v}) = 0 \quad \text{interchanging both sides,}$$

$$\vec{r} \times \vec{v} = \vec{h} \dots \dots \dots (iii)$$

Where $\vec{h}$ is the constant of integration.

Now, Dot multiply both sides by $\vec{r}$ in equation (iii)

$$\vec{r} \cdot (\vec{r} \times \vec{v}) = \vec{r} \cdot \vec{h}$$

$$\Rightarrow 0 = \vec{r} \cdot \vec{h} \quad \left(\because \vec{a} \cdot (\vec{a} \times \vec{b}) = 0 \right)$$

$$\text{or } \vec{r} \cdot \vec{h} = 0$$

This shows that the position vector of the particle at any time is perpendicular to the fixed constant vector $\vec{h}$ and hence lies in a plane.

Lecture (35)

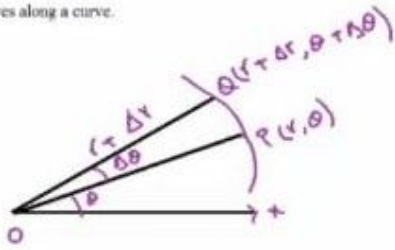
Orbital Motion Theorem 2

Theorem:

When a particle moves under a central force, the areal velocity is constant.

Where areal velocity is the rate at which area is swept out by a particle as it moves along a curve.

moves along a curve.



Solution:

$$-F = ma$$

$$-F = m[(\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{s}]$$

$$= m[(\ddot{r} - r\dot{\theta}^2)\hat{r} + \frac{1}{r}(2r\dot{r}\dot{\theta} + r^2\ddot{\theta})\hat{s}]$$

$$-F = m[(\ddot{r} - r\dot{\theta}^2)\hat{r} + \frac{1}{r}\frac{d}{dt}(r^2\dot{\theta})\hat{s}]$$

$$-F_r\hat{r} - F_\theta\hat{s} = \dots\dots\dots$$

$$-F_r = m(\ddot{r} - r\dot{\theta}^2)$$

$$-F_\theta = \frac{m}{r}\left(\frac{d}{dt}(r^2\dot{\theta})\right)$$

$$F_\theta = 0$$

$$\frac{m}{r}\frac{d}{dt}(r^2\dot{\theta}) = 0$$

$$\Rightarrow \frac{d}{dt}(r^2\dot{\theta}) = 0$$

$$\Rightarrow r^2\dot{\theta} = \text{const} = h$$

$$\Delta A = \text{Area of } OP\theta \cong \text{Area of } OPR$$

$$= \frac{1}{2}OQ \times PR$$

$$= \frac{1}{2}(r + \Delta r)r \sin \Delta\theta$$

$$= \frac{1}{2}(r^2 \sin \Delta\theta + r.\Delta r \sin \Delta\theta)$$

$$\frac{\Delta A}{\Delta t} = \frac{1}{2}\left(r^2 \frac{\Delta\theta}{\Delta t} + r.\Delta r.\frac{\Delta\theta}{\Delta t}\right)$$

$$\text{to having limit as } \Delta t \rightarrow 0, \Delta r \rightarrow 0, \Delta\theta \rightarrow 0$$

$$\frac{\Delta A}{\Delta t} = \frac{1}{2}r^2\dot{\theta} + r.\theta.\dot{\theta}$$

$$= \frac{1}{2}r^2\dot{\theta}$$

$$= \frac{1}{2}h$$

$$2\left(\frac{dA}{dt}\right) = h$$

$$\Rightarrow h \text{ is equal to twice the Areal speed.}$$

Lecture (36)

Differential Equation of the Orbital in Pedal Form

As we know the differential equation of the orbit of a particle in polar form is

$$\frac{d^2u}{d\theta^2} + u = \frac{F}{h^2u^2} \dots\dots\dots(i)$$

Also, we have $\frac{1}{p^2} = u^2 + \left(\frac{du}{d\theta}\right)^2$ Differentiating w.r. 'θ',

$$-2p^{-3} \cdot \frac{dp}{d\theta} = 2u \cdot \frac{du}{d\theta} + 2\left(\frac{du}{d\theta}\right)\left(\frac{d^2u}{d\theta^2}\right)$$

$$\Rightarrow -\frac{1}{p^3} \cdot \frac{dp}{d\theta} = u \frac{du}{d\theta} + \frac{du}{d\theta} \cdot \frac{d^2u}{d\theta^2}$$

or $-\frac{1}{p^3} \cdot \frac{dp}{d\theta} \cdot \frac{du}{d\theta} = \frac{du}{d\theta} \left(u + \frac{d^2u}{d\theta^2}\right)$

$$\Rightarrow -\frac{1}{p^3} \cdot \frac{dp}{du} = \frac{d^2u}{d\theta^2} + u \quad \text{using equation (i)}$$

$$-\frac{1}{p^3} \cdot \frac{dp}{du} = \frac{F}{h^2u^2}$$

$$-\frac{1}{p^3} \cdot \frac{dp}{dr} \cdot \frac{dr}{du} = \frac{F}{h^2u^2} \dots\dots\dots(ii) \quad \text{But } r = \frac{1}{u} \Rightarrow u = \frac{1}{r}$$

$$\Rightarrow \frac{dr}{du} = -\frac{1}{u^2} = -r^2$$

Equation (ii) becomes $-\frac{1}{p^3} \cdot \frac{dp}{dr} \cdot (-r^2) = \frac{F}{h^2\left(\frac{1}{r^2}\right)}$

$$\Rightarrow \frac{h^2}{p^3} \cdot \frac{dp}{dr} = F$$

This is the differential equation of the orbit of a particle in pedal form.

Lecture (37)

Apse and Apsidal Distance

Apse:

An apse is a point on an orbit at which the tangent to the orbit is perpendicular to the radius

vector drawn from the centre of force. An apse is shown in the figure.

Apsidal Distance:

The distance of an apse from the centre of force is called apsidal distance.

Apse line:

The line joining an apse to the centre of force is called apse line.

Apsidal Angle:

The angle between two consecutive apse line is called an apsidal angle.

The Condition for An Apside:

As we know that the angle ϕ between the tangent to the orbit at a point

$P(r, \theta)$

and radius vector $\vec{r}$ is given by $\tan \phi = r \frac{d\theta}{dr}$, for an apse, $\phi = \frac{\pi}{2}$

$$\text{So, } \tan \frac{\pi}{2} = r \frac{d\theta}{dr} \Rightarrow \infty = r \frac{d\theta}{dr}$$

$$\Rightarrow \frac{1}{r} \cdot \frac{dr}{d\theta} = 0$$

$$\text{So, } \frac{dr}{d\theta} = 0, \dots\dots\dots (i) \text{ is the required condition}$$

$$\text{Also, } r = \frac{1}{u}$$

$$\Rightarrow \frac{dr}{d\theta} = -\frac{1}{u^2} \cdot \frac{du}{d\theta}$$

$$\text{Equation (i) becomes } -\frac{1}{u^2} \cdot \frac{du}{d\theta} = 0$$

$$\text{So, } \frac{du}{d\theta} = 0, \dots\dots\dots (ii)$$

equation (i) and (ii) are the required conditions.

Lecture (38)

Classification of Orbits in Term of Energy

Theorem:

Show that the orbit described by the planet around sun is a conic.

The planet will described an ellipse, a parabola or a hyperbola according as total energy per unit area is negative , zero or positive.

Proof:

We consider the motion of the planet round the sun and the force is governed by Newton's Law of Gravitation. If 'M' and 'm' are the mass of sun and the planet then they attract each other with a force MmG/r^2 where G is constant of gravitation. Take the sun as the pole, the differential equation of the orbit is

$$\frac{d^2u}{d\theta^2} + u = \frac{F}{h^2u^2} \dots\dots\dots(1)$$

$$F = \frac{MmG}{r^2}$$

$$\frac{d^2u}{d\theta^2} + u = \frac{MmG/r^2}{h^2u^2}$$

$$h^2u^2 \left(\frac{d^2u}{d\theta^2} + u \right) = MmG/r^2$$

$$h^2u^2 \left(\frac{d^2u}{d\theta^2} + u \right) = \mu \times u^2 \qquad MmG = \mu$$

$$\frac{d^2u}{d\theta^2} + u = \frac{\mu}{h^2}$$

$$u = \frac{1}{r}$$

$$u^2 = \frac{1}{r^2}$$

$$\frac{d^2u}{d\theta^2} + u - \frac{\mu}{h^2} = 0$$

$$\frac{d^2}{d\theta^2} \left(u - \frac{\mu}{h^2} \right) + \left(u - \frac{\mu}{h^2} \right) = 0$$

$$(D^2 + 1) \left(u - \frac{\mu}{h^2} \right) = 0$$

$$m^2 + 1 = 0 \quad \Rightarrow m = \pm 1$$

$$u - \frac{\mu}{h^2} = A \cos \theta + B \sin \theta$$

$$u - \frac{\mu}{h^2} = C \cos \theta_e \cos \theta + C \sin \theta_e \sin \theta$$

$$u = \frac{\mu}{h^2} + C \cos(\theta - \theta_e)$$

$$\theta_e = 0$$

$$u = \frac{\mu}{h^2} + C \cos \theta \qquad u = \frac{1}{r}$$

$$\frac{1}{r} = \frac{\mu}{h^2} \left[1 + \frac{C}{\frac{\mu}{h^2}} \cos \theta \right]$$

$$\frac{h^2}{ur} = 1 + \frac{h^2 C}{\mu} \cos \theta \dots\dots\dots(2)$$

$$\frac{h^2/\mu}{r} = 1 + \frac{h^2 c}{\mu} \cos \theta.$$

$$\frac{l}{r} = 1 + e \cos \theta$$

$$l = h^2/\mu \qquad e = \frac{h^2 c}{\mu}$$

$$u = \frac{\mu}{h^2} + \cos \theta \dots\dots\dots(A)$$

$$Total \ Energy = K.E + P.E$$

$$E = T + V$$

$$V = -\int F(r) dr$$

$$= -\int -\frac{\mu}{h^2} dr$$

Lecture (39)

Kepler's Laws of Planetary Motion

- 1) Each planet describes an ellipse with sun as focus.
- 2) The areal speed of the radius vector in any orbit is constant.
- 3) The square of time period for describing the whole orbit is proportional to the

cube of the major axis of this orbit.

1ST LAW OF KEPLER'S

Each planet describes an ellipse with sun as focus.

Proof:

Proof:
According to Newton's law of gravitation if M and m denote the masses of sun and planet respectively at a distance ' r ' apart, they attract each other with a force $\frac{GMm}{r^2}$.

So, the equation of the orbit of planet in polar form referred to sun as pole is

$$mh^2u^2\left(\frac{d^2u}{d\theta^2} + u\right) = \frac{GMm}{r^2}$$

$$\Rightarrow h^2u^2\left(\frac{d^2u}{d\theta^2} + u\right) = GMu^2 \quad \left(\because \frac{1}{r} = u\right)$$

Let $GM = \mu$

So,

$$h^2u^2\left(\frac{d^2u}{d\theta^2} + u\right) = \mu u^2$$

$$\frac{d^2u}{d\theta^2} + u = \frac{\mu}{h^2}$$

or

$$\frac{d^2u}{d\theta^2} + \left(u - \frac{\mu}{h^2}\right) = 0 \dots\dots\dots (i)$$

Let

$$u - \frac{\mu}{h^2} = V \dots\dots\dots (ii)$$

$$\Rightarrow \frac{du}{d\theta} = \frac{dV}{d\theta} \Rightarrow \frac{d^2u}{d\theta^2} = \frac{d^2V}{d\theta^2}$$

$\therefore$ Equation (i) becomes

$$\frac{d^2V}{d\theta^2} + V = 0$$

$$\Rightarrow (D^2 + 1)V = 0 \quad \text{where } D = \frac{d}{d\theta}$$

The auxiliary equation is $n^2 + 1 = 0 \Rightarrow n = \pm i$

So,

$$V = C_1 \cos\theta + C_2 \sin\theta$$

Put $C_1 = A \cos\alpha$ and $C_2 = A \sin\alpha$ (where A and α are constant)

$$V = A \cos\alpha \cos\theta + A \sin\alpha \sin\theta$$

$$\Rightarrow V = A \cos(\theta - \alpha)$$

If we rotate the initial line about/through an angle α , the above equation becomes

$$V = A \cos\theta$$

$\therefore$ Equation (ii) becomes $u - \frac{\mu}{h^2} = A \cos\theta$

$$\Rightarrow uh^2 - \mu = A h^2 \cos\theta$$

or

$$\frac{h^2u}{\mu} = \frac{A h^2 \cos\theta}{\mu} + 1$$

$$\Rightarrow \frac{h^2u}{\mu} = \frac{A h^2}{\mu} \cos\theta + 1$$

which represents an ellipse of the form $\frac{\ell}{r} = e \cos\theta + 1$

where ℓ is semi-latusrectum and e is the eccentricity. Here

$$\ell = \frac{h^2}{\mu}, \text{ and } e = \frac{A h^2}{\mu} = A \ell$$

AREAL VELOCITY:

Areal Velocity:

The area swept out by the radius vector per unit time of a particle moving in a central orbit, is called the areal velocity of the particle.

2nd LAW OF KEPLER'S:

The areal speed of the radius vector in any orbit is constant.

The areal speed of the radius vector in any orbit is constant.

Proof:

Let the sun is located at origin 'O' and 'P' be the position of the planet at any time 't' and

$$\vec{OP} = \vec{r}$$

Let after a very small interval of time δt

planet is at point Q and $\vec{OQ} = \vec{r} + \delta \vec{r}$

$$\vec{PQ} = \vec{r} + \delta \vec{r} - \vec{r}$$

$$\Rightarrow \vec{PQ} = \delta \vec{r}$$

Let δA be the sectorial area OPQ.

$$\text{Now, } \delta A = \frac{1}{2} |\vec{OP} \times \vec{OQ}|$$

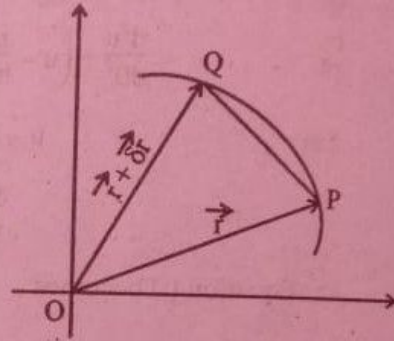
$$\delta A = \frac{1}{2} |\vec{r} \times \delta \vec{r}| \quad \text{Dividing both sides by } \delta t \neq 0, \text{ we get}$$

$$\frac{\delta A}{\delta t} = \frac{1}{2} \left| \vec{r} \times \frac{\delta \vec{r}}{\delta t} \right| \quad \text{Taking limit } \delta t \rightarrow 0,$$

$$\lim_{\delta t \rightarrow 0} \frac{\delta A}{\delta t} = \lim_{\delta t \rightarrow 0} \frac{1}{2} \left| \vec{r} \times \frac{\delta \vec{r}}{\delta t} \right|$$

$$\Rightarrow \frac{dA}{dt} = \frac{1}{2} \left| \vec{r} \times \frac{d\vec{r}}{dt} \right|$$

$$\text{or} \quad \frac{dA}{dt} = \frac{1}{2} |\vec{r} \times \vec{v}| \quad \left(\because \frac{d\vec{r}}{dt} = \vec{v} \right)$$



$$\Rightarrow \frac{dA}{dt} = \frac{1}{2} |\vec{h}| \quad (\because \vec{r} \times \vec{v} = \vec{h})$$

So, $\frac{dA}{dt} = \frac{1}{2} h$

Since h is constant, So $\frac{dA}{dt}$ is also constant i.e., the area swept out by the radius vector per unit time remains constant.

3RD LAW OF KEPLER'S:

Square of the time period for describing whole orbit is proportional to the cube of major axis of this orbit.

Proof:

vector per unit time remains constant.

3rd LAW OF KEPLER'S

Square of the time period for describing whole orbit is proportional to the cube of major axis of this orbit.

Proof: If a and b are semi axis of the elliptic orbit described by a planet the time for

$$\text{one revolution} = T = \frac{\pi ab}{\text{areal velocity}} = n \frac{\pi ab}{\frac{h}{2}}$$

$$= \frac{2\pi ab}{h}$$

$$\text{Now } h^2 = \mu l, \quad h = \sqrt{\mu l}, \text{ But } l = \frac{b^2}{a}$$

$$h = \sqrt{\frac{\mu b^2}{a}} = \frac{\sqrt{\mu}}{\sqrt{a}} b$$

$$\therefore T = \frac{2\pi ab}{\frac{\sqrt{\mu}}{\sqrt{a}} b} = \frac{2\pi a^{3/2}}{\sqrt{\mu}}$$

$$T^2 = \frac{4\pi^2 a^3}{\mu}$$

$$= \frac{\pi^2}{2\mu} \cdot 8a^3$$

$$= \frac{\pi^2}{2\mu} (2a)^3$$

$$\Rightarrow T^2 \propto (2a)^3$$

Example:

Example 1: Show that the law of force towards the pole, of a particle describing the curve $r^n = a^n \cos n\theta$ is given by $f = \frac{(n+1)h^2 a^{2n}}{r^{2n+3}}$.

Solution:

As we know $f = h^2 u^2 \left(\frac{d^2 u}{d\theta^2} + u \right) \dots\dots\dots (i)$

Now, $r^n = a^n \cos n\theta$

$\Rightarrow \frac{1}{u^n} = a^n \cos n\theta \quad \left(\because r = \frac{1}{u} \right)$

$\Rightarrow u^n = \frac{1}{a^n} \sec n\theta$

Differentiating both sides w.r.t. ' θ ', $nu^{n-1} \frac{du}{d\theta} = \frac{1}{a^n} (\sec n\theta \tan n\theta)(n)$

$\Rightarrow \frac{du}{d\theta} = \frac{1}{a^n u^{n-1}} (\sec n\theta \tan n\theta)$

or $\frac{du}{d\theta} = \frac{u}{a^n u^n} (\sec n\theta \tan n\theta)$

But $u^n = \frac{1}{a^n} \sec n\theta \dots\dots\dots (A)$

So, $\frac{du}{d\theta} = \frac{1}{a^n \left(\frac{\sec n\theta}{a^n} \right)} (\sec n\theta \tan n\theta)$

$\Rightarrow \frac{du}{d\theta} = u \tan n\theta \dots\dots\dots (ii)$

Differentiating again w.r.t. ' θ ', $\frac{d^2 u}{d\theta^2} = \frac{du}{d\theta} \tan n\theta + u \sec^2 n\theta (n)$

Using equation (ii), $\frac{d^2 u}{d\theta^2} = u \tan n\theta \cdot \tan n\theta + u n \sec^2 n\theta$

$\Rightarrow \frac{d^2 u}{d\theta^2} = u (\tan^2 n\theta + n \sec^2 n\theta) \dots\dots\dots (iii)$

Put (iii) in equation (i),

$f = h^2 u^2 [u (\tan^2 n\theta + n \sec^2 n\theta) + u]$

$\Rightarrow f = h^2 u^3 (\tan^2 n\theta + n \sec^2 n\theta + 1)$

$\Rightarrow f = h^2 u^3 (\sec^2 n\theta + n \sec^2 n\theta)$

$\Rightarrow f = h^2 u^3 (n+1) \sec^2 n\theta$

From equation (A), $\sec n\theta = a^n u^n$

So, $f = h^2 u^3 (n+1) a^{2n} u^{2n}$

ORBITAL MOTION

$\Rightarrow$

So,

$$f = h^2 a^{2n} (n+1) u^{2n+3}$$

$$f = \frac{(n+1) h^2 a^{2n}}{r^{2n+3}} \quad \left(\because u = \frac{1}{r} \right)$$

Lecture (40, 41) include

(Example of Orbital Motion), (Example of Orbital Motion II)

Example:

Q.4. A particle of unit mass describes an ellipse under the action of central force Mr. Show that the normal component of the acceleration at any instant is $\frac{abM^{3/2}}{V}$, where V is the velocity at that instant and a, b are semi-axis of ellipse.

Solution:

Since, the particle of unit mass describes an ellipse under the action of central force Mr. So the equation of its orbit is

$$h^2 u^2 \left(\frac{d^2 u}{d\theta^2} + u \right) = Mr$$

$$\Rightarrow h^2 u^2 \left(\frac{d^2 u}{d\theta^2} + u \right) = \frac{M}{u} \quad \left(\because r = \frac{1}{u} \right)$$

$$\Rightarrow h^2 \left(\frac{d^2 u}{d\theta^2} + u \right) = \frac{M}{u^3} \quad \text{Multiplying both sides by } 2 \frac{du}{d\theta}$$

$$h^2 \left(2 \frac{du}{d\theta} \frac{d^2 u}{d\theta^2} + 2u \frac{du}{d\theta} \right) = 2Mu^{-3} \frac{du}{d\theta} \quad \text{Integrating both sides,}$$

$$h^2 \left[\left(\frac{du}{d\theta} \right)^2 + u^2 \right] = 2Mu^{-2} + C$$

$$\Rightarrow h^2 \left[\left(\frac{du}{d\theta} \right)^2 + u^2 \right] = -\frac{M}{u^2} + C$$

$$\text{But } \left(\frac{du}{d\theta} \right)^2 + u^2 = \frac{1}{p^2}$$

So,

$$\frac{h^2}{p^2} = -\frac{M}{u^2} + C$$

or

$$\frac{h^2}{p^2} = C - Mr^2 \quad \dots\dots\dots (i)$$

$$\text{Now, pedal equation of ellipse is } \frac{a^2 b^2}{p^2} = a^2 + b^2 - r^2 \quad \dots\dots\dots (ii)$$

Comparing equation (i) with equation (ii), we have

$$\frac{h^2}{a^2 b^2} = \frac{C}{a^2 + b^2} = \frac{M}{1}$$

$$h^2 = Ma^2 b^2 \quad \dots\dots\dots (iii)$$

Now,

$$Vp = h$$

$$V^2 p^2 = h^2$$

$$V^2 p^2 = Ma^2 b^2$$

[using equation (iii)]

$$\Rightarrow p^2 = \frac{Ma^2b^2}{V^2}$$

Now, for ellipse, we have

$$\rho = \frac{a^2b^2}{p^2}$$

The normal component of acceleration is

$$a_n = \frac{V^2}{\rho} = \frac{V^2}{\frac{a^2b^2}{p^2}}$$

$$\text{So, } a_n = \frac{V^2 p^3}{a^2 b^2} \dots \dots \dots \text{(iv)}$$

$$\text{Now, } p^2 = \frac{Ma^2b^2}{V^2} \quad \text{(Taking square root)}$$

$$\Rightarrow p = \frac{M^{1/2}ab}{V} \quad \text{(Taking cube)}$$

$$\Rightarrow p^3 = \frac{M^{3/2}a^3b^3}{V^3}$$

$$\text{Put this value in equation (iv), } a_n = \frac{V^2}{a^2b^2} \cdot \frac{M^{3/2}a^3b^3}{V^3}$$

$$\text{Hence, } a_n = \frac{abM^{3/2}}{V} \text{ as required.}$$

Example:

Q.10. A planet is describing an ellipse about the sun as focus. Show that its velocity away from the sun is greatest when the radius vector to the planet is at right angles to the major axis of the path, and that it then is $\frac{2\pi a e}{T(1-e^2)}$, where $2a$ the major axis, e the eccentricity and T is the periodic time.

Solution:

Since the orbit of the planet is an ellipse with force center at one of the foci, so its equation is

$$r = \frac{\ell}{1 + e \cos \theta} \quad \text{or} \quad \frac{\ell}{r} = 1 + e \cos \theta$$

$$\Rightarrow r = \ell(1 + e \cos \theta)^{-1}$$

$$\Rightarrow \frac{dr}{d\theta} = -\ell(1 + e \cos \theta)^{-2} (-e \sin \theta)$$

$$\frac{dr}{d\theta} = \frac{e\ell \sin \theta}{(1 + e \cos \theta)^2}$$

Now,
$$\frac{dr}{dt} = \frac{dr}{d\theta} \cdot \frac{d\theta}{dt}$$

$$\Rightarrow \frac{dr}{dt} = \frac{e\ell \sin \theta}{(1 + e \cos \theta)^2} \cdot \dot{\theta} \quad \left(\because \frac{d\theta}{dt} = \dot{\theta} \right) \quad \text{But } 1 + e \cos \theta = \frac{\ell}{r}$$

$$\text{So, } \frac{dr}{dt} = \frac{e/\sin\theta}{\frac{\ell^2}{r^2}} \dot{\theta}$$

$$\frac{dr}{dt} = \frac{e \sin\theta}{\ell} \cdot (r^2 \dot{\theta}) \quad \text{But } r^2 \dot{\theta} = h$$

$$\text{So, } \frac{dr}{dt} = \frac{eh \sin\theta}{\ell} \quad \text{But } h^2 = \mu\ell \Rightarrow h = \sqrt{\mu\ell}$$

$$\text{So, } \frac{dr}{dt} = \frac{e \sin\theta}{\ell} \cdot \sqrt{\mu\ell}$$

$$\Rightarrow \frac{dr}{dt} = e \sin\theta \sqrt{\frac{\mu}{\ell}}$$

where e , μ and ℓ are constants. So, $\frac{dr}{dt}$ is maximum if $\sin\theta$ is maximum.

$$\text{i.e., } \sin\theta = 1$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

$$\text{So, } \left(\frac{dr}{dt}\right)_{\max} = e \frac{\sqrt{\mu}}{\sqrt{\ell}} \dots\dots\dots (i)$$

$$\text{But, by Kepler's law, } T = \frac{2\pi a^{3/2}}{\sqrt{\mu}} \Rightarrow \sqrt{\mu} = \frac{2\pi a^{3/2}}{T} \dots\dots\dots (ii)$$

$$\text{Also, } \ell = \frac{b^2}{a} \Rightarrow \ell = \frac{a^2(1-e^2)}{a}$$

$$\ell = a(1-e^2) \dots\dots\dots (iii)$$

Put (ii), (iii) values in equation (i), we get

$$\left(\frac{dr}{dt}\right)_{\max} = e \cdot \frac{2\pi a^{3/2}}{T} \cdot \frac{1}{\sqrt{a(1-e^2)}}$$

$$\Rightarrow \dot{r} = \frac{2\pi ae}{T\sqrt{1-e^2}} \text{ as required.}$$

Collision of Particles

Direct Collision:

Two bodies are said to be Impinge (collide) directly when the direction of each is along

the common normal at the point of contact.

Oblique Collision:

When the direction of motion either or both is not along the common normal.

This impact is said to be oblique.

Elastic Collision:

In an Elastic collision K.E is the same before and after collision.

Inelastic Collision:

If the K.E. before and after collision is not same then the collision is said to be inelastic collision.

Newton's Law Experimental View or Newton's Law of Restitution

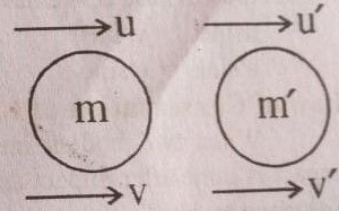
Case a:

Newton's Law Experimental View or Newton's Law of Restitution

CASE a: When two bodies impinge directly, their relative velocity after impact is in a constant ratio to their relative velocity before impact and is in the opposite direction.

Thus if the velocities of two impinging bodies be u, u'

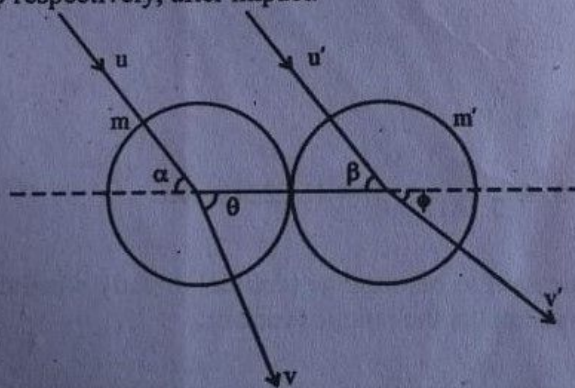
before impact and v, v' after impact, then $\frac{v - v'}{u - u'}$ is negative and is equal to a constant quantity which is independent of the masses m and m' of the bodies. This constant quantity is denoted by " e " is called the modulus or coefficient of Elasticity or restitution or resilience of the bodies



$$\frac{v - v'}{u - u'} = -e \quad \Rightarrow \quad v - v' = -e(u - u')$$

CASE b: When two bodies impinge obliquely their relative velocity resolved along their common normal after impact is in constant ratio to their relative velocity before impact resolved in the same direction and is in the opposite direction.

Thus if two bodies of masses m, m' be moving with velocities u, u' in direction inclined at angle α, β respectively to their common normal and v, v' be their velocities in direction θ and ϕ respectively, after impact.



$$\frac{v \cos \theta - v' \cos \phi}{u \cos \alpha - u' \cos \beta} = -e$$

$$v \cos \theta - v' \cos \phi = -e(u \cos \alpha - u' \cos \beta)$$

Motion Perpendicular to The Line of Impact:

Motion Perpendicular To The Line of Impact:

When two smooth bodies impinge, there is no tangential action between them. The action and reaction between them is entirely along the common normal. Therefore there is no force perpendicular to the line of impact and consequently no change of velocity in that direction.

Hence the resolved parts of velocity of each body perpendicular to the common normal i.e. along tangent at the point of contact before impact is equal to its resolved parts the impact thus in case b, we have;

$$u \sin \alpha = v \sin \theta$$

$$u' \sin \beta = v' \sin \phi$$

Law of Conservation of Linear Momentum:

When two bodies impinge the sum of their momentums along the line of impact is same after impact as before;

For case a: $mv + m'v' = mu + m'u'$

For case b: $mv \cos \theta + m'v' \cos \phi = mu \cos \alpha + m'u' \cos \beta$

Direct Collision Between A Particle And Fixed Barrier:

Direct Collision Between A Particle And Fixed Barrier:

Let a particle of mass 'm' is moving with a velocity 'u' and it collides with a fixed barrier let $\bar{v}$ is the velocity after impact. Let the barrier supply impulse on the body which reversed its motion. Then impulse is given by;

$$I = mv - m(-u) = mv + mu$$

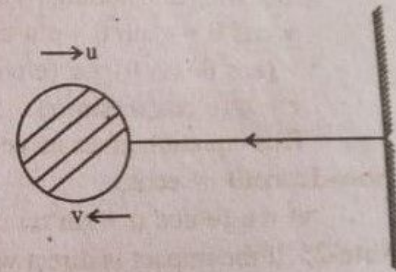
$$I = m(v+u) \dots \dots \dots (i)$$

The loss in energy can be analyzed by using newton's law of restitution.

$$-e = \frac{v - 0}{-u - 0} \Rightarrow e = \frac{v}{-u}$$

$$e = \frac{v}{u} \Rightarrow v = eu$$

Hence the sphere impinges directly on a smooth plane, it rebounds the reverse direction with velocity eu.



Lecture (43)

Damped Harmonic Oscillator

The Harmonic Oscillator:

A mathematical model describing problems of oscillations in the form of rectilinear simple

Harmonic motion is usually called for Harmonic oscillator e.g. simple pendulum

Damped Harmonic Oscillator:

When some sort of velocity dependent retarding (or damping) force, due to resistance of the

medium or some other possible cause is ordinary present when the damping force is taken into account. The harmonic oscillator is termed as the Damped Harmonic Oscillator.

Damped Oscillation under the Damping Force Proportional (to the 1st power of) Velocity:

Suppose a particle of mass m is moving along the x -axis having fixed origin O .

Let P be the

position of the particle at time t from O . The forces acting on the particle are

- (i) a restorative force $-mw^2 x_i$ and
(ii) a damping force $-mk\dot{x}_i$, where k is +ve constant.

The equation of motion is $m\ddot{x}_i = -mw^2 x_i - mk\dot{x}_i$

or $\ddot{x} + k\dot{x} + w^2 x = 0 \dots\dots\dots (i)$

set $D = \frac{d}{dt} \therefore (1) \text{ becomes}$

$$(D^2 + kD + w^2)x = 0$$

if $k = 0$ and $w \neq 0$, this gives S.H.M. if $k \neq 0$ then the auxiliary equation is

$$\text{Hence } p = \frac{-k \pm \sqrt{k^2 - 4w^2}}{2} = \frac{-k}{2} \pm \sqrt{\frac{k^2}{4} - w^2}$$

Now we discuss the three cases for the solution of different equation according as

$$\text{the disc} = \frac{k^2}{4} - w^2 \leq 0$$

Case I:

$$\text{when } \frac{k^2}{4} - w^2 < 0$$

$$\frac{k^2}{4} - w^2 = -n^2$$

$$p = -\frac{k}{2} \pm n$$

Therefore the general solution of different equation is

$$x = e^{\frac{-k}{2}t} [C_1 \cos nt + C_2 \sin nt]$$

put $C_1 = a \cos \alpha$, where C_1, C_2, a, α are constant we get

$$x = ae^{\frac{-k}{2}t} [\sin \alpha \cos nt + \cos \alpha \sin nt]$$

$$x = ae^{\frac{-k}{2}t} \sin(nt + \alpha) \dots\dots\dots (ii) \quad \text{as } t \rightarrow \infty, x \rightarrow 0$$

Case II:

$$\text{when } \frac{k^2}{4} - w^2 > 0 \text{ Let } \frac{k^2}{4} - w^2 = \lambda^2$$

$$p = -\frac{k}{2} \pm \lambda$$

$$\text{Therefore the equation is } x = Ae^{\left(\frac{-k}{2} + \lambda\right)t} + Be^{\left(\frac{-k}{2} - \lambda\right)t}$$

as $t \rightarrow \infty, x \rightarrow 0$

Case III:

$$\text{when } \frac{k^2}{4} - w^2 = 0 \quad \text{then } p = -\frac{k}{2}, -\frac{k}{2} \quad \text{Let } \frac{k}{2} = w$$

$$\therefore p = -w, -w$$

$\therefore$ The solution of different equation is

$$x = (A + Bt)e^{-wt} \text{ as } t \rightarrow \infty, x \rightarrow 0$$

Lecture (44)

Damped Forced Oscillations

Damped Forced Oscillations:

In order to overcome the damping effect of a medium an applied force called driving force is generally applied, In such a case the system is called damped forced oscillator.

The applied force is usually a periodic force, In addition to the restorative force and damping force. Let we apply a force $mF \cos pt$, (F and p being constants) acting in the direction of increasing x

Then equation of motion is $m\ddot{x} = -mw^2x - mk\dot{x} + mF \cos pt$ dividing by m

$$\begin{aligned} \ddot{x} + k\dot{x} + w^2x &= F \cos pt \\ \Rightarrow (D^2 + kD + w^2)x &= F \cos pt \dots\dots\dots(i) \end{aligned}$$

The solution of this Differentiate equation consists of two parts the complementary function and the particular integral solution. First we find the complimentary solution of $(D^2 + kD + w^2)x = 0$

In the above three cases we see that the complimentary function decay with time. For this reason this part of the solution is called the transient solution. The particular integral on the other hand generally does not die out, therefore the ultimate motion of the system depends upon that part of the solution which is called the steady state solution:

$$\text{P.I } x_p = \frac{1}{D^2 + kD + w^2} F \cos pt$$

$$x_p = F \cdot \frac{1}{-p^2 + kD + w^2} \cos pt$$

$$= F \cdot \frac{1}{kD + (w^2 - p^2)} \cos pt$$

$$= F \frac{kD - (w^2 - p^2)}{k^2 D^2 - (w^2 - p^2)^2} \cos pt$$

$$= \frac{F \cdot [kD - (w^2 - p^2)]}{-k^2 p^2 - (w^2 - p^2)^2} \cdot \cos pt$$

$$= F \cdot \frac{(w^2 - p^2) - kD}{k^2 p^2 + (w^2 - p^2)^2} \cos pt$$

$$= \frac{F (w^2 - p^2) \cos pt + kp \sin pt}{(w^2 - p^2)^2 + k^2 p^2}$$

$$\text{Let } \frac{(w^2 - p^2)}{R} = \sin \alpha, \frac{kp}{R} = \cos \alpha \quad R^2 = (w^2 - p^2)^2 + k^2 p^2$$

$$= \frac{F[R \sin \alpha \cos pt + R \cos \alpha \sin pt]}{R^2}$$

$$= \frac{F}{R^2} [R \sin \alpha \cos pt + R \cos \alpha \sin pt]$$

$$= \frac{F}{R} \sin(\alpha + pt) \text{ is the steady state solution.}$$

The steady state which is periodic function (of time) will not decay with time.

The amplitude of the steady state is $\frac{F}{R}$ and will always lies between $-\frac{F}{R}$ and $\frac{F}{R}$.

Lecture (45)

Vertical Motion with air Resistance

Case I: Vertically downward Motion:

CHAPTER 10

DAMPED AND FORCED OSCILLATOR RESISTED MOTION

AIR RESISTANCE PROPORTION TO THE SQUARE OF THE VELOCITY

Case 1: Vertically downward Motion:

To discuss the motion of a particle of mass m released from rest at an initial height above the ground. The resistance of air is assumed to be proportional to the square of velocity.

Let x be the distance fallen through in time t , with air retardation $m\lambda v^2$ where v be the velocity at time t and λ is +ve constant. Then equation of motion is $mv \frac{dv}{dx} = mg - m\lambda v^2$ divide by m both side

$$v \frac{dv}{dx} = g - \lambda v^2$$

$$v \frac{dv}{dx} = \lambda \left(\frac{g}{\lambda} - v^2 \right)$$

$$\text{Let } \frac{g}{\lambda} = k^2 \quad \Rightarrow \quad v \frac{dv}{dx} = \lambda (k^2 - v^2) \dots\dots\dots (i)$$

Separate the variable and integrating

$$\int \frac{v dv}{k^2 - v^2} = \lambda \int dx$$

$$-\frac{1}{2} \int \frac{2v dv}{k^2 - v^2} = \lambda x + C$$

$$-\frac{1}{2} \ln (k^2 - v^2) = \lambda x + C, \quad \text{Where } C \text{ is constant of integration}$$

Initially when $t = 0$, $x = 0$, $v = 0$

Putting these values in (i) we get $C = -\frac{1}{2} \ln k^2$

$$\text{Hence } -\frac{1}{2} \ln (k^2 - v^2) = \lambda x - \frac{1}{2} \ln k^2$$



$$\Rightarrow \frac{1}{2} \ln k^2 - \frac{1}{2} \ln (k^2 - v^2) = \lambda x$$

$$\Rightarrow -\frac{1}{2} \{ \ln (k^2 - v^2) - \ln k^2 \} = \lambda x$$

$$\Rightarrow \ln \left(\frac{k^2 - v^2}{k^2} \right) = -2\lambda x$$

$$\Rightarrow \frac{k^2 - v^2}{k^2} = e^{-2\lambda x} \Rightarrow 1 - \frac{v^2}{k^2} = e^{-2\lambda x}$$

$$\Rightarrow 1 - e^{-2\lambda x} = \frac{v^2}{k^2}$$

$$\Rightarrow v^2 = k^2(1 - e^{-2\lambda x}) \dots\dots\dots (ii)$$

This equation gives us velocity in terms of x .

Again use the expression for acceleration.

$$\frac{dv}{dt} = \lambda (k^2 - v^2) \text{ again separate the variables and integrating}$$

$$\int \frac{dv}{k^2 - v^2} = \lambda \int dt$$

$$\frac{1}{k} \tan^{-1} \frac{v}{k} = \lambda t + D$$

Initially when $t = 0$, $v = 0$

$$\frac{1}{k} \tan^{-1} (0) = \lambda(0) + D \Rightarrow D = 0 \text{ put in above}$$

$$\frac{1}{k} \tan^{-1} \frac{v}{k} = \lambda t \Rightarrow \tan^{-1} \frac{v}{k} = \lambda k t \text{ But } k^2 = \frac{g}{\lambda} \Rightarrow \lambda = \frac{g}{k^2}$$

$$\tan^{-1} \frac{v}{k} = \frac{g}{k^2} t$$

$$\Rightarrow \tan^{-1} \frac{v}{k} = \frac{g}{k^2} t$$

$$\frac{v}{k} = \tan^{-1} \frac{gt}{k} \Rightarrow v = k \tanh \left(\frac{gt}{k} \right) \text{ this gives 'v' in terms 't'}$$

$$v^2 = k^2 \tanh^2 \left(\frac{gt}{k} \right) \dots\dots\dots (iii)$$

Now we find the relation between x and t from (ii) and (iii) by comparing.

$$k^2 \tanh^2 \left(\frac{gt}{k} \right) = k^2(1 - e^{-2\lambda x})$$

$$\Rightarrow \tanh^2 \left(\frac{gt}{k} \right) = 1 - e^{-2\lambda x}$$

DAMPED AND FORCED OSCILLATOR RESISTED MOTION (CHAPTER 10)

$$e^{-2\lambda x} = 1 - \tanh^2\left(\frac{gt}{k}\right)$$

$$\Rightarrow e^{-2\lambda x} = \operatorname{sech}^2\left(\frac{gt}{k}\right)$$

$$e^{2\lambda x} = \cosh^2\left(\frac{gt}{k}\right)$$

$$\Rightarrow e^{\lambda x} = \cosh\left(\frac{gt}{k}\right)$$

$$\lambda x = \ln \cosh\left(\frac{gt}{k}\right)$$

$$\Rightarrow x = \frac{1}{\lambda} \ln \cosh\left(\frac{gt}{k}\right) \quad \text{But } \lambda = \frac{g}{k^2}, \Rightarrow \frac{1}{\lambda} = \frac{k^2}{g}$$

$$x = \frac{k^2}{g} \ln \cosh\left(\frac{gt}{k}\right)$$

Case II: Vertically Upward Motion

Case II: Vertically upward Motion:

Now we discuss the motions of a particle of mass m projected vertically upward with an initial velocity v_0 and air resistance proportional to the square of the velocity.

Let x be the height attained by the particle in time t . the equation

of motion is $m \frac{dv}{dx} = -mg - m\lambda v^2$ Divide by m

$$v \frac{dv}{dx} = -g - \lambda v^2 = -\lambda \left(\frac{g}{\lambda} + v^2 \right) \text{ Let } \frac{g}{\lambda} = k^2$$

$$v \frac{dv}{dx} = -\lambda(k^2 + v^2) \dots\dots (i) \text{ Separate the variables and integrating}$$

$$\int \frac{v dv}{k^2 + v^2} = -\int \lambda dx \quad \text{Multiply both side by 2}$$

$$\int \frac{2v dv}{k^2 + v^2} = -2\lambda \int dx$$

$$\ln(k^2 + v^2) = -2\lambda x + A \quad \text{where } A \text{ is constant of integration.}$$

Initially when $t = 0, x = 0,$

$v = v_0$ we have

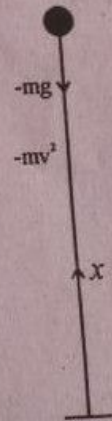
$$A = \ln(k^2 + v_0^2) \quad \text{put in above}$$

$$\ln(k^2 + v^2) = -2\lambda x + \ln(k^2 + v_0^2)$$

$$\ln(k^2 + v^2) - \ln(k^2 + v_0^2) = -2\lambda x$$

$$\ln \left(\frac{k^2 + v^2}{k^2 + v_0^2} \right) = -2\lambda x$$

$$\Rightarrow x = \frac{1}{2\lambda} \ln \left(\frac{k^2 + v_0^2}{k^2 + v^2} \right) \quad \text{But } \lambda = \frac{g}{k^2} \text{ Or } \frac{1}{\lambda} = \frac{k^2}{g}$$



$$x = \frac{k^2}{2g} h \left(\frac{k^2 + v_0^2}{k^2 + v^2} \right) \dots\dots\dots (ii)$$

Again (i) can be written as

$$\frac{dv}{dt} = -\lambda (k^2 + v^2) \quad \text{Separate the variables and integrating}$$

$$\int \frac{dv}{k^2 + v^2} = -\lambda \int dt$$

$$\frac{1}{k} \tan^{-1} \frac{v}{k} = -\lambda t + B$$

At $t = 0$, $v = v_0$,

$$B = \frac{1}{k} \tan^{-1} \frac{v_0}{k}$$

Above become

$$\frac{1}{k} \tan^{-1} \frac{v}{k} = -\lambda t + \frac{1}{k} \tan^{-1} \frac{v_0}{k}$$

$$\frac{1}{k} \tan^{-1} \frac{v_0}{k} - \frac{1}{k} \tan^{-1} \frac{v}{k} = \lambda t$$

$$\tan^{-1} \frac{v_0}{k} - \tan^{-1} \frac{v}{k} = \lambda t k \quad \text{But } \lambda = \frac{g}{k^2}$$

$$\tan^{-1} \frac{v_0}{k} - \tan^{-1} \frac{v}{k} = t k \frac{g}{k^2}$$

$$\tan^{-1} \frac{v_0}{k} - \tan^{-1} \frac{v}{k} = \frac{gt}{k} \dots\dots\dots (iii)$$

$$\tan^{-1} \frac{v}{k} = \tan^{-1} \frac{v_0}{k} - \frac{gt}{k}$$

$$\frac{v}{k} = \tan \left(\tan^{-1} \frac{v_0}{k} - \frac{gt}{k} \right)$$

$$v = k \tan \left(\tan^{-1} \frac{v_0}{k} - \frac{gt}{k} \right)$$

$$\frac{dx}{dt} = k \tan \left(\tan^{-1} \frac{v_0}{k} - \frac{gt}{k} \right) \quad \text{Separate the variable and integrating}$$

$$\int dx = k \int \tan \left(\tan^{-1} \frac{v_0}{k} - \frac{gt}{k} \right) dt$$

$$x = -k \ln \cos \left(\tan^{-1} \frac{v_0}{k} - \frac{gt}{k} \right) \left(\frac{k}{g} \right) + C$$

$$x = \frac{k^2}{g} \ln \cos \left(\tan^{-1} \frac{v_0}{k} - \frac{gt}{k} \right) + C$$

at $t = 0, x = 0$, so $C = -\frac{k^2}{g} \ln \cos \left(\tan^{-1} \frac{v_0}{k} \right)$

above become

$$x = \frac{k^2}{g} \ln \cos \left(\tan^{-1} \frac{v_0}{k} - \frac{gt}{k} \right) - \frac{k^2}{g} \ln \cos \left(\tan^{-1} \frac{v_0}{k} \right)$$

$$= \frac{k^2}{g} \ln \left[\frac{\cos \left(\tan^{-1} \frac{v_0}{k} - \frac{gt}{k} \right)}{\cos \left(\tan^{-1} \frac{v_0}{k} \right)} \right] \dots \dots \dots (iv)$$

Equation (ii), (iii) and (iv) determine the motion.

Best of luck & Pray for me